

RCEP and Supply Chain Relationships of Chinese Firms: Evidence from Relationship-Pair Data

Anonymous manuscript

Abstract

International supply chains depend on durable firm-to-firm relationships, yet it remains unclear whether regional trade agreements alter these links. The Regional Comprehensive Economic Partnership (RCEP), which entered into force in 2022, provides a setting to test whether regional trade rules reshape cross-border relationships. Year-end activity captures the extensive margin of relationship organization: retaining or establishing a link requires search, coordination, and switching costs that trade values alone do not reveal. Using FactSet Revere data, we construct a balanced panel of 630,280 relationship-year observations for 78,785 customer and supplier pairs involving 5,689 Chinese listed firms from 2017–2024. A difference-in-differences design compares RCEP-partner relationships with other foreign-partner relationships before and after entry, using relationship-pair and year fixed effects. RCEP is associated with a 1.36 percentage-point increase in the probability that a relationship is active at year-end. The response is concentrated in supplier links, which rise by 2.95 percentage points, while customer links show no discernible change. Country-level evidence is consistent with tariff relief, deeper intermediate-goods integration, and lower dependence on individual partner sources as channels. These findings show that regional agreements can reshape micro-level supply-chain links and highlight relationship continuity for sourcing strategies and trade-policy design.

Keywords: RCEP, supply chains, relationship-pair panel, difference-in-differences, trade policy

1. Introduction

Economic activity in Asia is increasingly organized through dense cross-border production networks in which firms repeatedly source inputs from, and

sell output to, partners in neighboring economies. Recent evidence points to considerable persistence in the region’s aggregate trade network even after a large common shock (Kiyota, 2022), while new measures of regional value-chain position emphasize the importance of tracing production links beyond gross trade flows (He et al., 2023). Whether a firm–partner link is active at year-end is economically meaningful because it captures the extensive margin of cross-border organization. Forming or maintaining a relationship requires search, screening, coordination, and sometimes relationship-specific investment, while ending one can entail switching costs and disrupt future trade. These organizational margins are not observed in trade values conditional on a shipment. The entry into force of the Regional Comprehensive Economic Partnership (RCEP) on January 1, 2022 began bringing 15 Asia-Pacific economies under a common trade framework. Its combination of coordinated tariff schedules, regional cumulation of origin, and trade-facilitation commitments can change both the cost of a cross-border transaction and the set of partners with which that transaction is feasible. This institutional change therefore raises a more specific economic question: does RCEP affect whether a particular cross-border customer or supplier relationship is active at year-end after entry into force?

Firms are the primary organizational units through which production, sourcing, and sales are coordinated, so the consequences of regional integration should ultimately appear in their cross-border commercial ties. We study directed relationships between Chinese listed firms and foreign partners. A customer relationship is a disclosed link in which the foreign partner purchases from the Chinese firm, whereas a supplier relationship is a link in which the foreign partner supplies the Chinese firm. Using relationship-level data from the FactSet Revere database, we construct a balanced annual panel of 630,280 observations for 78,785 directed firm–partner pairs. The empirical design compares relationships with partners in RCEP member economies to relationships with other foreign partners before and after the common 2022 treatment date. The specification includes firm–partner–direction pair fixed effects and year fixed effects, with standard errors clustered by relationship pair.

The main estimate is 0.0136 (standard error 0.0040), implying a 1.36 percentage-point differential increase in the probability that a relationship is active at year-end. The estimate is 0.0144 (standard error 0.0041) when member-specific entry dates are used and remains positive in several alternative time windows and sample exclusions. The response is concentrated

in supplier relationships, for which the estimated increase is 2.95 percentage points; the corresponding estimate for customer relationships is small and statistically insignificant. At the partner-country-year level, RCEP exposure is associated with tariff relief, a higher intermediate-goods share in China’s imports from member economies, and a lower partner-country share of China’s imports. These aggregate trade patterns, together with the supplier-side relationship response, are consistent with stronger regional production-network integration helping Chinese firms maintain or establish disclosed supplier relationships. The joint pre-treatment Wald test does not reject at the 5% level but does reject at the 10% level ($p = 0.078$). We therefore interpret the estimates as a positive relative response while keeping the limitations of both the timing evidence and the comparison group explicit.

The paper is positioned against several closely related bodies of work. Research on trade agreements has established effects on bilateral trade, firm outcomes, and third-country adjustment (Baier and Bergstrand, 2007; Sun et al., 2022; Lee et al., 2023), while recent RCEP studies evaluate regional value chains in a GVC-CGE framework (Sun et al., 2023) or examine how financing constraints relate to manufacturers’ export partnerships with RCEP economies (Peng et al., 2025). A second literature studies firms’ use of preferences and the sourcing consequences of rules of origin (Hayakawa et al., 2023; Tanaka and Fukunishi, 2022; Head et al., 2024). A third analyzes how trade-policy changes propagate through supplier search and firm-to-firm production links (Garg et al., 2023; Grossman et al., 2024). These studies motivate a relationship-level test but do not establish whether RCEP’s entry into force altered the annual activity of directed customer and supplier links.

Relative to this work, the paper makes three contributions. First, it shifts the outcome from aggregate trade or export participation to the annual activity of disclosed firm-partner relationships and estimates the policy response within the same directed pair over time. Second, it relates partner-country-year changes in tariff relief, the composition of intermediate-goods imports, and partner-country import shares to relationship activity, while recognizing that these aggregate measures do not observe firm-level preference claims, certificates of origin, customs transactions, or relationship values. Third, it separates supplier from customer links and examines how the initial relationship response varies with pre-policy relationship maturity, prior trade-agreement coverage, and manufacturing exposure.

The remainder of the paper is organized as follows. Section 2 reviews the literature and develops the hypotheses. Section 3 describes the sample,

variables, and empirical design. Section 4 presents the baseline estimates, mechanism analysis, and robustness tests. Section 5 examines heterogeneous responses. Section 6 provides further evidence on a competing explanation. Section 7 concludes with policy implications.

2. Literature review and hypothesis development

2.1. Literature review

The first strand of related research examines how trade agreements and trade institutions affect bilateral trade and firm behavior. Gravity-based evidence establishes that free-trade agreements increase trade between members (Baier and Bergstrand, 2007). Recent JAE studies extend the analysis from trade flows to export-product quality and foreign direct investment, examine environmental provisions as a determinant of firm-level quality adjustment, and estimate employment responses to regional integration (Sun et al., 2022; Narayan et al., 2022; Zhu and Sun, 2025; Li et al., 2025). The incidence of an agreement nevertheless depends on the institutional details faced by firms. Hayakawa et al. (2023), for example, measure the firm-level utilization of regional trade agreements from the importer’s perspective, while Ni et al. (2022) analyze heterogeneous trade effects across sectors. Other recent work examines how tariff costs and trade-policy uncertainty shape cross-border organization, export behavior, and input choice (Xie, 2023; Zhou and Wen, 2022; Handley et al., 2024; Poilly and Tripier, 2025). Firm-level evidence on third-country effects further shows why adjustment outside an agreement must be part of the comparison (Lee et al., 2023). This literature implies that membership and statutory tariff schedules alone are insufficient measures of firms’ effective exposure.

The second strand studies regional production networks and RCEP more directly. Recent work constructs measures of regional dual value chains (He et al., 2023), evaluates RCEP through a GVC-CGE model (Sun et al., 2023), and examines the reorganization of export-product structures as global value chains change (Zhang and Ding, 2024). Complementary JAE studies examine domestic value added in exports, GVC upgrading after foreign-capital withdrawal, financial and institutional determinants of value-chain participation, and risk-related changes in regional trade structure (Du et al., 2023; Wu et al., 2023; Xu et al., 2024; Ali et al., 2026; Liu et al., 2026). At the firm level, Peng et al. (2025) study financing constraints and export

partnerships with RCEP economies. Other evidence connects lower cross-border mobility costs to bilateral value-chain linkages (Ma et al., 2024b) and digital development to economies’ positions in the intermediate-goods trade network (Huang et al., 2026). Work on data restrictions and digital-trade provisions further illustrates that contemporary regional integration operates through non-tariff rules as well as customs duties (Ma et al., 2024a; Li and Li, 2026). These contributions establish the relevance of regional integration for production networks, but aggregate value-chain measures, simulations, and exporter-country links do not reveal whether a particular disclosed customer or supplier relationship remains active after implementation.

The third strand concerns rules of origin and the organization of input sourcing. Rules of origin determine whether firms can claim preferences and can therefore either promote regional sourcing or impose compliance costs. Tanaka and Fukunishi (2022) find that relaxing origin requirements produced heterogeneous export responses across developing-economy beneficiaries. Studying reforms in Japan, Hayakawa and Yamanouchi (2023) document an increase in imports from least-developed economies but heterogeneous effects on domestic producers. Theoretically, Head et al. (2024) show that tighter content requirements initially expand regional parts sourcing but reduce it once the requirement passes a threshold. The design of these rules is itself endogenous to negotiation and differs between intermediate and final products (Hayakawa et al., 2024); overlapping preference regimes add another layer to firms’ compliance choices (Hayakawa, 2022). Recent research on rules of origin likewise links these provisions to imported inputs and firm innovation (Zhou and Wu, 2025). Taken together, this work makes input sourcing central to the expected effect of RCEP’s regional-cumulation provision.

A final set of studies highlights that supply chains are networks of durable relationships rather than anonymous spot transactions. Administrative shipment data reveal that ownership boundaries, distance, relationship specificity, and input characteristics shape where firms source products (Garg et al., 2023). Models of unexpected tariffs similarly predict renegotiation with incumbent suppliers or costly search for replacement partners (Grossman et al., 2024). Empirical studies connect imported inputs and supply-chain links to firm productivity (Newman et al., 2023), show that shocks can propagate through observed firm networks (Crosignani et al., 2022), and examine supply-chain spillovers in corporate innovation (Ou et al., 2026). The present paper joins this relationship-based perspective to the RCEP literature. Its

empirical unit preserves the firm, partner, and relationship direction, allowing the analysis to distinguish changes in relationship activity from changes in trade values or aggregate network position.

2.2. Hypothesis development

RCEP was signed on November 15, 2020 and entered into force for the first group of members on January 1, 2022. The agreement covers China, Japan, South Korea, Australia, New Zealand, and the ten ASEAN economies. It combines market-access commitments with rules on origin, customs procedures, services, investment, intellectual property, and electronic commerce (Shen and Li, 2020; Ping, 2021). Recent research on digital-trade rules in regional agreements underscores the network implications of this broader institutional scope (Li and Li, 2026). RCEP’s common origin regime is especially relevant in Asia, where firms previously navigated several overlapping ASEAN-plus-one agreements whose product-level rules reflected different negotiating interests (Hayakawa, 2022; Hayakawa et al., 2024).

The tariff schedule is phased. Approximately 65 percent of tariff lines receive immediate elimination, while other lines are reduced over five to twenty years and a small set is excluded. More than 90 percent of tariff lines are scheduled eventually to reach zero, but the timing differs by product and member (Liu and Zhen, 2022). The 2022–2024 observation window therefore captures the initial institutional response and only the early portion of tariff liberalization; it should not be read as an estimate of the agreement’s full long-run tariff effect.

The value of a supply-chain relationship reflects more than the current tariff on a shipment. Establishing a link requires partner search, screening, negotiation, and adaptation to relationship-specific production needs. Once formed, the link remains active when the expected surplus from continued trade exceeds the cost of renegotiation or replacement. This logic is consistent with evidence that relationship specificity and input characteristics shape firms’ sourcing choices (Garg et al., 2023) and with models in which tariff shocks induce renegotiation with incumbent suppliers or costly search for alternatives (Grossman et al., 2024).

RCEP raises the expected surplus from qualifying intraregional trade through phased tariff reductions, common procedures, and regional cumulation. Value added from any member can count toward the originating status of a product, subject to the agreement’s regional-value-content rules. The provision broadens the set of production arrangements capable of receiving

preferential treatment and reduces the need to satisfy separate origin systems for each bilateral route. Utilization is not automatic—firms must still find compliance worthwhile (Hayakawa et al., 2023)—but the joint reduction in trade and coordination costs can both sustain an existing link and make a newly disclosed link worthwhile. This reasoning yields the first hypothesis:

H1. *RCEP’s entry into force increases the probability that a disclosed supply-chain relationship between a Chinese listed firm and a partner in an RCEP economy is active.*

The agreement can sustain relationships through three related channels. First, incremental tariff relief raises the net return to transactions with partners for which RCEP improves on the pre-existing tariff schedule. This margin should be strongest where the additional preference is economically meaningful, but its effect depends on actual utilization rather than the statutory rate alone. Second, regional cumulation increases the value of intermediate inputs obtained within the agreement because content contributed by one member can help a downstream product qualify in another. Evidence that origin rules affect export responses (Tanaka and Fukunishi, 2022; Hayakawa and Yamanouchi, 2023) and theoretical results linking content requirements to regional parts sourcing (Head et al., 2024) support this input-reallocation channel. Recent firm-level work that connects origin rules with imported inputs provides a closely related motivation (Zhou and Wu, 2025).

Third, a common origin regime can change the concentration of eligible sourcing. Under separate bilateral agreements, a firm may organize production around a narrow set of partners that satisfy route-specific requirements. Cumulation makes inputs from multiple member economies mutually usable for origin purposes, widening the feasible regional supplier set and reducing dependence on any one partner source. Research on import diversification under a regional cooperation initiative provides a related application of this sourcing margin (Zhu et al., 2025). This prediction does not require every firm to diversify immediately: origin rules may generate nonlinear responses (Head et al., 2024), and reforms have produced heterogeneous effects across products and firms (Tanaka and Fukunishi, 2022; Hayakawa and Yamanouchi, 2023). It instead implies that, on average, tariff relief, stronger intermediate-goods trade integration among members, and a lower partner-country import share are observable implications of the proposed channels. These trade measures are evaluated at the partner-country–year level. Their relationship-level implication is that deeper regional production integration can make supplier ties more valuable to maintain or establish. These arguments yield the second

hypothesis:

H2a. *RCEP exposure is associated with greater incremental tariff relief, which is positively related to relationship activity.*

H2b. *RCEP exposure is associated with a larger intermediate-goods share in bilateral imports, which is positively related to relationship activity.*

H2c. *RCEP exposure is associated with a lower partner-country share of China’s imports, and this reduction is positively related to relationship activity.*

The relevance of these channels should differ by relationship direction. A supplier relationship records a foreign partner providing inputs or products to the Chinese firm, whereas a customer relationship records sales by the Chinese firm to a foreign partner. Tariff preferences can affect both directions, but regional cumulation operates first through the origin, eligibility, and organization of inputs. The supplier side is therefore exposed directly to the decision margins emphasized by research on imported-input choice (Handley et al., 2024), within-firm and external sourcing (Garg et al., 2023), and the productivity role of imports transmitted through supply chains (Newman et al., 2023). Customer relationships may respond only after changes in input composition, production costs, or market access pass through to downstream sales. Over the first three years of implementation, this ordering implies a larger and more readily detectable response among supplier links. We therefore propose the third hypothesis:

H3a. *The increase in relationship activity following RCEP is stronger for supplier relationships than for customer relationships.*

Relationship-specific capital may also shape the scope for adjustment. Long-standing links have already absorbed search, screening, negotiation, and adaptation costs, leaving less room for an institutional change to alter annual activity. This reasoning implies:

H3b. *The increase in relationship activity following RCEP is larger for relationships that were not long-standing before policy implementation.*

RCEP did not provide the same degree of institutional novelty across members. China already had free-trade agreements with the ASEAN members, Australia, New Zealand, and Korea, whereas RCEP created the first free-trade agreement between China and Japan. The expected response should therefore be larger where the agreement introduced genuinely new bilateral rules:

H3c. *The increase in relationship activity following RCEP is stronger for Japanese partners than for partners already covered by an earlier agreement*

with China.

Finally, regional cumulation and origin requirements are most directly relevant to firms that organize manufacturing inputs across borders. This gives rise to the final heterogeneity hypothesis:

H3d. *Among supplier relationships, the increase in relationship activity following RCEP is stronger for manufacturing firms than for non-manufacturing firms.*

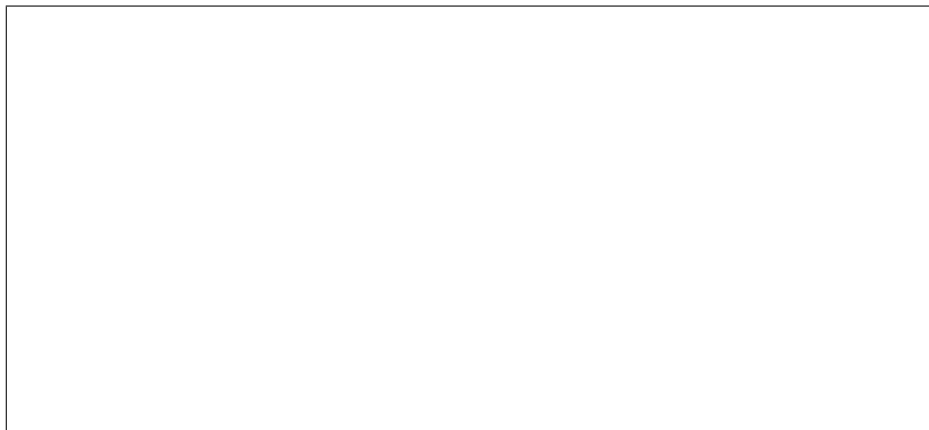


Figure 1: Research framework

3. Research design

3.1. Sample and data

The relationship data are from FactSet Revere’s Supply Chain Relationships database. We retain CUSTOMER and SUPPLIER records involving a Chinese listed firm and a foreign partner, and preserve the direction of the link. The cleaned source contains 578,142 eligible disclosure records for 2017–2024. Multiple disclosure episodes involving the same Chinese firm, partner, and direction are consolidated into 78,785 distinct relationship pairs. Expanding each pair over the eight calendar years yields the balanced panel of 630,280 relationship–year observations used in the baseline analysis.

Formally, let $\mathcal{R}$ denote the set of directed firm–partner relationships observed at least once in the disclosure data and let $\mathcal{T}$ denote the calendar years 2017–2024. The analysis sample is the Cartesian product

$$\mathcal{P} = \mathcal{R} \times \mathcal{T} = \{(i, j, r, t) : (i, j, r) \in \mathcal{R}, t \in \mathcal{T}\}, \quad (1)$$

which preserves the firm, partner, and direction of each link while allowing its activity status to vary by year.

Firm financial information is from the China Stock Market and Accounting Research (CSMAR) database. We retain annual consolidated statements, exclude financial firms and ST firms, and winsorize continuous firm variables at the first and 99th percentiles. The matched firm-year file contains 35,978 observations for 5,689 firms (including 3,643 manufacturing firms).

Tariff data combine most-favored-nation rates with official additional-duty schedules from a major non-RCEP import market. Product-level rates are mapped from HS8 products to two-digit national industry categories using a 5,724-row concordance. The industry exposure series uses fixed 2016 Chinese export weights. For the further analysis, firm-level exposure is fixed in 2019 and equals the export-weighted average of the applicable additional duties. Neither measure uses post-policy trade shares.

3.2. Variable selection and description

3.2.1. Supply-chain relationship activity

The unit of observation is a Chinese firm–foreign partner–direction pair observed at the end of each calendar year. Let i index the Chinese firm, j the foreign partner, and r the direction (supplier or customer). The outcome, $Active_{ijrt}$, equals one if at least one disclosure episode for pair (i, j, r) has started by December 31 of year t and has not ended by that date; it is zero otherwise. Thus, the panel records the extensive margin of relationship activity among pairs observed at least once in the source data, rather than the probability of forming every possible firm combination.

If s_{ijre} and e_{ijre} denote the start and end dates of disclosure episode e (with an ongoing episode assigned an end date after the sample), this outcome can be written as

$$Active_{ijrt} = \mathbb{1}\{\exists e : s_{ijre} \leq \text{Dec. 31}(t) < e_{ijre}\}. \quad (2)$$

3.2.2. RCEP treatment

The treatment is $Post2022_t \times RCEP_j$. In the common-date specification, $Post2022_t$ equals one for calendar years 2022–2024 and zero for 2017–2021. $RCEP_j$ equals one when the partner is located in one of the 14 RCEP economies other than China. This specification treats 2022, when the agreement first entered into force, as the common policy date. A sensitivity specification instead assigns 2023 to Indonesia and the Philippines, whose implementation dates were later. The relationship-pair fixed effect, α_{ijr} , absorbs

all time-invariant characteristics of a disclosed pair, while year fixed effects absorb common annual shocks.

Thus, the common-date treatment indicator used in the baseline model is

$$D_{jt} = RCEP_j \times Post2022_t, \quad Post2022_t = \mathbb{1}\{t \geq 2022\}. \quad (3)$$

3.2.3. Covariates and auxiliary measures

Table 1 reports the variables used in the analysis. The firm covariates are the log of total assets (“Size”), book leverage (“Lev”), return on assets (“ROA”), annual revenue growth (“Growth”), and inventory divided by total assets (“InvDensity”). They are defined at the firm-year level; the preferred fixed-effects estimate is intentionally identified from the treatment interaction and pair and year fixed effects, while these covariates enter the high-dimensional nuisance models in the double-machine-learning sensitivity analysis. The pre-policy relationship-age measure is the number of years from the first disclosed start date to December 31, 2021 for pairs active in 2021. Pre-policy sourcing breadth counts the distinct RCEP partner countries connected to a Chinese firm through supplier links in 2021; the large-firm indicator is based on the median of firms’ 2017–2021 mean log assets. Manufacturing status is based on the firm’s latest available primary-industry classification observed no later than 2021.

The policy-linked indicators are defined at the partner-country-year level and merged to every relationship involving that country. Incremental tariff relief is the 2017–2019 import-weighted scheduled tariff reduction under the relevant RCEP schedule. The intermediate-goods share is China’s imports of goods classified as intermediate inputs under the strict BEC5 definition from partner country j , divided by China’s total imports from that partner in year t . It measures the composition of bilateral imports and, across RCEP members, the degree of regional intermediate-goods trade integration; it is not a measure of firm i ’s input purchases. The partner-country import share is country j ’s share of China’s total imports in year t ; a lower value indicates less dependence on that individual source. These aggregates describe policy-relevant conditions and do not measure a firm’s preference claim, certificate of origin, transaction value, or overall firm-level sourcing concentration.

3.3. Model specification

The empirical design exploits the common entry-into-force date while allowing exposure to vary across partner economies. We compare the evolution

of relationships involving RCEP partners with that of relationships involving other foreign partners. Pair fixed effects remove time-invariant differences across disclosed firm–partner–direction pairs, and year fixed effects remove shocks common to all relationships. Identification therefore comes from the differential post-2022 change between the two partner groups. The identifying condition is that, absent RCEP, their relationship activity would have followed parallel trends. It also requires the comparison group not to experience differential spillovers from the agreement. That requirement may be imperfect because RCEP can redirect sourcing and sales between member and non-member economies: trade diversion that reduces non-member relationships would enlarge the estimated contrast, whereas positive spillovers to non-members would attenuate it. The coefficient should therefore be interpreted as the post-entry change for RCEP relationships relative to non-RCEP relationships, rather than as an isolated effect against a completely untreated counterfactual. We estimate a linear probability model because the outcome is an indicator and interpret the coefficient as a percentage-point change.

The baseline specification is

$$Active_{ijrt} = \beta(Post2022_t \times RCEP_j) + \alpha_{ijr} + \lambda_t + \varepsilon_{ijrt}, \quad (4)$$

where α_{ijr} denotes firm–partner–direction pair fixed effects and λ_t denotes year fixed effects. The coefficient β is the differential change in the probability that an observed relationship is active for RCEP partners relative to other foreign partners. Standard errors are clustered by relationship pair, matching the level at which serially correlated relationship outcomes and treatment histories are observed.

3.4. Descriptive statistics

Table 1: Summary statistics

Variable	Observations	Mean	S.D.	Min.	Median	Max.
<i>Relationship-year variables</i>						
Active relationship	630,280	0.226	0.418	–	–	–
RCEP partner	630,280	0.183	0.386	–	–	–
Post-2022 indicator	630,280	0.375	0.484	–	–	–
RCEP $\times$ Post-2022	630,280	0.068	0.253	–	–	–
Foreign supplier relationship	630,280	0.413	0.492	–	–	–
Relationship age in 2021 (years)	630,280	0.478	1.387	0.000	0.000	18.746
Pre-policy RCEP sourcing breadth	630,280	0.512	1.302	0.000	0.000	8.000
Large-firm indicator	362,440	0.501	0.500	–	–	–
<i>Partner-country-year indicators</i>						
Incremental tariff relief (percentage points)	1,368	0.157	2.490	0.000	0.000	65.751
Intermediate-goods share of bilateral imports (percent)	1,355	66.811	28.800	0.000	73.156	100.000
Partner-country share of China's imports (percent)	1,376	0.579	1.469	0.000	0.035	10.882
<i>Firm-year covariates</i>						
Log total assets	35,978	22.213	1.426	19.142	22.021	26.588
Leverage	35,978	0.411	0.206	0.056	0.399	0.924
Return on assets	35,978	0.032	0.074	-0.309	0.036	0.213
Revenue growth	30,283	0.103	0.329	-0.613	0.067	1.731
Inventory intensity	35,337	0.128	0.111	0.000	0.104	0.609

Notes: Statistics cover 2017–2024 and use all available observations; counts vary with missing values. Binary variables report the mean and S.D. only, with – indicating omitted distribution statistics. Percentage measures use the units shown in the variable labels.

Table 1 shows that the balanced relationship-year panel contains 630,280 observations. The smaller counts for the partner-country-year indicators and firm covariates reflect source-file coverage and missing values rather than changes to the baseline sample. Figure 2 then traces the unconditional active-relationship rates for the two partner groups.

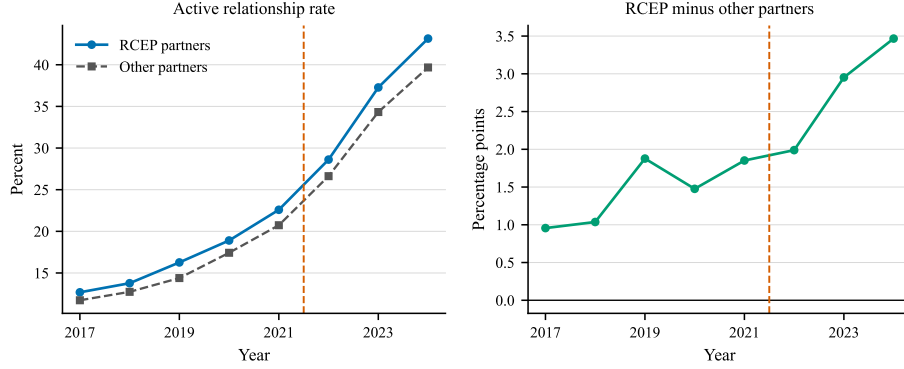


Figure 2: Descriptive relationship activity by partner group, 2017–2024

Notes: The left plot reports the share of relationship pairs that are active at each year-end, separately for RCEP and other foreign partners. The right plot reports the difference between the two rates in percentage points. The vertical line marks the January 2022 entry into force. The figure is descriptive and does not condition on the fixed effects used in the regressions.

Active rates rise for both groups over the sample. The rate for RCEP-partner pairs increases from 12.68 percent in 2017 to 43.14 percent in 2024, while the corresponding rate for other foreign partners rises from 11.72 to 39.68 percent. The raw gap therefore moves from 0.95 percentage points in 2017 to 3.47 percentage points in 2024, with a 1.85-point gap immediately before entry into force. These unconditional patterns motivate the difference-in-differences comparison but are not themselves evidence of a causal effect.

4. Empirical results and robustness tests

4.1. RCEP and supply-chain relationship activity

Table 2 presents the baseline estimates. The preferred coefficient is 0.0136 with a standard error of 0.0040 ($p = 0.001$), corresponding to a 1.36 percentage-point differential increase in the active-relationship probability. Using member-specific entry dates gives 0.0144 (standard error 0.0041), a nearly identical estimate. The stability of the point estimate indicates that imposing a common 2022 treatment date is not responsible for the baseline result. The positive and statistically significant estimates are consistent with H1.

Table 2: Baseline results

	(1) Common 2022 date Pair-clustered S.E.	(2) Member-specific dates Pair-clustered S.E.
RCEP treatment	0.0136*** (0.0040)	0.0144*** (0.0041)
Observations	630,280	630,280
Relationship pairs	78,785	78,785
Pair fixed effects	Yes	Yes
Year fixed effects	Yes	Yes

Notes: The outcome is an active firm–partner relationship at year-end. All columns include pair and year fixed effects. Column (1) uses a common 2022 entry date; column (2) uses member-specific entry dates. Standard errors are clustered by relationship pair and reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.2. Mechanism analysis

The theoretical discussion identifies three policy-linked channels associated with relationship activity: incremental tariff relief, regional intermediate-goods trade integration, and reduced dependence on an individual partner-country source. Each measure varies at the partner-country–year level and is assigned to relationship (i, j, r) according to the partner’s country. Denoting the relevant country-level measure by M_{jt} , we estimate

$$M_{jt} = a(Post2022_t \times RCEP_j) + \alpha_{ijr} + \lambda_t + u_{ijrt}, \quad (5)$$

$$Active_{ijrt} = c'(Post2022_t \times RCEP_j) + bM_{jt} + \alpha_{ijr} + \lambda_t + v_{ijrt}. \quad (6)$$

Equations (5) and (6) define the two paths. For each indicator, the total effect c is re-estimated from Eq. (4) on the same complete-case sample used in the two path equations; this is why the reported values of c differ slightly across rows and from the full-sample estimate in Table 2. The indirect association is ab , and the linear decomposition satisfies $c = c' + ab$. Inference for ab is based on 2,000 bootstrap samples drawn at the relationship-pair level; both path equations are re-estimated in every draw. This procedure preserves the panel structure and uses the same fixed effects and clustering level as the corresponding regressions.

For each mechanism, the total and direct paths are related to the indirect association by the linear decomposition

$$c = c' + ab, \quad (7)$$

where a is the first-stage treatment coefficient and b is the mediator coefficient in the outcome equation.

Table 3 reports the estimates. The percentile 95% bootstrap confidence intervals exclude zero for all three indirect associations: [0.0087, 0.0209] for incremental tariff relief, [0.0002, 0.0004] for the intermediate-goods share of bilateral imports, and [0.0240, 0.0307] for the partner-country import share. RCEP exposure is associated with a 0.0176 increase in the intermediate-goods share, equivalent to 1.76 percentage points. The corresponding indirect association with relationship activity is 0.0003, or 0.03 percentage points. This estimate indicates stronger intermediate-goods trade integration among RCEP members, not increased input purchases by any particular firm. Combined with the supplier-side estimate reported below, the result is consistent with regional production-network integration helping Chinese firms maintain or establish disclosed supplier relationships. RCEP exposure is also associated with a lower partner-country import share, and a lower share is in turn associated with more active relationships. In this third decomposition, however, the positive indirect association ($ab = 0.0274$) is offset by a negative direct coefficient ($c' = -0.0141$); it should not be interpreted as a proportion of the total effect. Overall, the three patterns are consistent with H2a–H2c as policy-linked channel evidence, but do not identify firm-level preference use, certification, or transaction-level adjustment.

Table 3: Policy-linked mechanism evidence

Mechanism measure	a	b	c	c'	Indirect effect (ab)	Bootstrap 95% CI
Incremental tariff relief	0.2186*** (0.0021)	0.0688*** (0.0148)	0.0133*** (0.0040)	-0.0017 (0.0050)	0.0150***	[0.0087, 0.0209]
Intermediate-goods import share	0.0176*** (0.0006)	0.0181*** (0.0035)	0.0151*** (0.0041)	0.0148*** (0.0041)	0.0003***	[0.0002, 0.0004]
Partner-country import share	-0.0127*** (0.0001)	-2.1607*** (0.1394)	0.0133*** (0.0040)	-0.0141*** (0.0043)	0.0274***	[0.0240, 0.0307]

Notes: Each row uses its mechanism-specific complete-case sample. The percentile interval for ab is based on 2,000 relationship-pair bootstrap draws. All models include pair and year fixed effects; standard errors in parentheses are clustered by pair. *, **, and *** denote 10%, 5%, and 1% significance, using the corresponding bootstrap intervals for ab . All mechanism measures vary at the partner-country–year level.

4.3. Robustness tests

4.3.1. Event-study test

Figure 3 plots the event-study coefficients relative to the year before entry into force. We test the null hypothesis that the four pre-treatment coefficients for 2017–2020 are jointly equal to zero. The Wald statistic is 8.389 with four degrees of freedom ($p = 0.078$). The null is therefore not rejected at the 5% level but is rejected at the 10% level. Accordingly, the event study provides only marginal support for the parallel-trends assumption rather than a definitive validation of it.

The dynamic specification replaces the single post indicator with event-time indicators:

$$Active_{ijrt} = \sum_{\substack{k \in \mathcal{K} \\ k \neq -1}} \beta_k (RCEP_j \times \mathbb{I}\{t - 2022 = k\}) + \alpha_{ijr} + \lambda_t + \varepsilon_{ijrt}, \quad (8)$$

where $k = -1$ (2021) is the omitted event time. The pre-treatment coefficients are jointly tested using a Wald restriction.

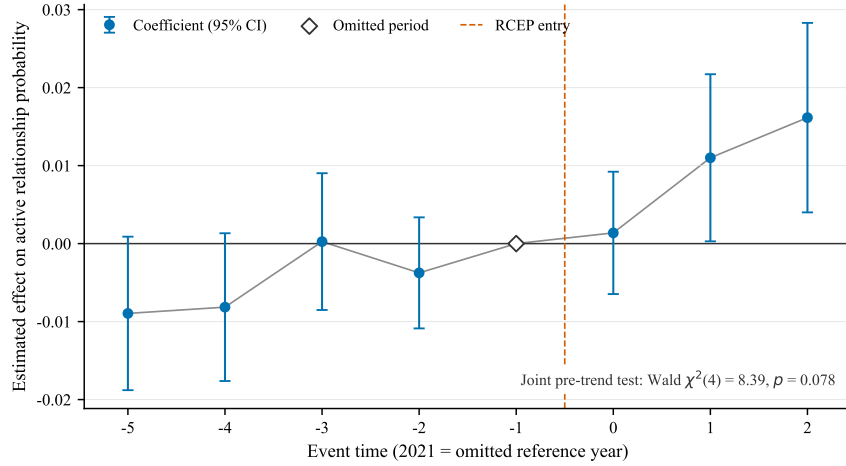


Figure 3: Event-study estimates of the RCEP relationship effect

Notes: Points are event-study coefficients from the relationship-pair fixed-effects model with year fixed effects; bars are 95 percent confidence intervals based on relationship-pair clustered standard errors. Event time -1 (2021) is the omitted period. The dashed vertical line marks RCEP's entry into force. The joint Wald test evaluates whether all four pre-treatment coefficients are zero.

4.3.2. Placebo tests

We next examine whether the estimated relationship is reproduced when either the policy timing or the country-level treatment assignment is falsified. The timing placebos assign the RCEP partner group to the 2020 and 2021 pre-entry calendar cutoffs. These dates correspond, respectively, to the agreement’s signing and the pre-implementation period, and therefore provide tests against anticipation or generic pre-entry changes. The spatial placebo randomly relabels 14 partner countries as treated in each of 499 draws, matching the number of RCEP economies in the data. The mixed placebo changes both the set of treated countries and the false cutoff; its joint randomization addresses the possibility that a particular country composition and a particular year jointly drive the estimate. These exercises are diagnostic: they test whether a comparable design generates a systematic positive coefficient in the absence of the actual policy assignment.

The timing estimates are statistically indistinguishable from zero for both pre-entry cutoffs: the coefficient is 0.0037 (standard error 0.0037, $p = 0.310$) for 2020 and 0.0051 (standard error 0.0040, $p = 0.199$) for 2021. In the country-label exercise, the 14-country placebo distribution has a mean of -0.0103 and a standard deviation of 0.0350; its two-sided reference probability is 0.774. The mixed country–date draws likewise remain centered close to zero, with a mean of -0.0056 , a standard deviation of 0.0160, and a reference probability of 0.407 for the 14-country setting. Figure 4 displays the country-label distribution, and Table 4 reports the selected timing, spatial, and mixed placebo estimates. Taken together, these falsification exercises do not reveal a systematic positive shift generated by arbitrary timing or country labels.

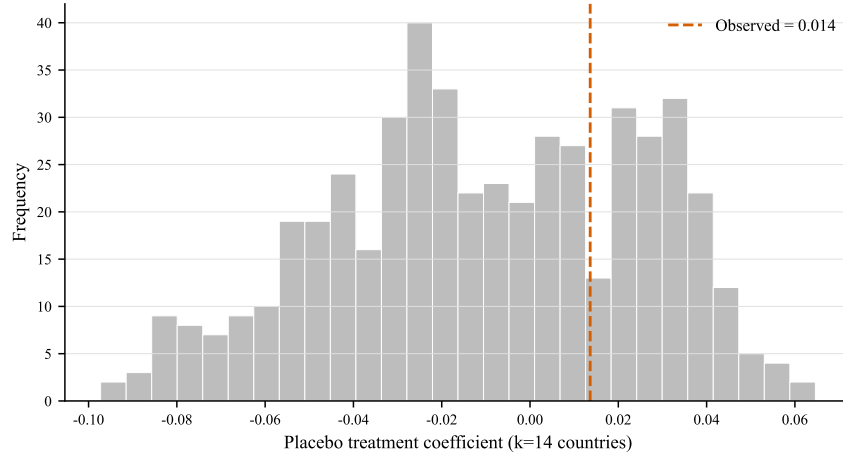


Figure 4: Country-label placebo distribution

Notes: The histogram reports 499 estimates obtained after assigning the treatment label to 14 randomly selected partner countries. The dashed line marks the estimate for the actual RCEP assignment. All estimates use the relationship-pair and year fixed-effects specification.

Table 4: Placebo tests

4em 0pt				
Placebo design	Setting	Estimate/mean	S.E./S.D.	Reference p
Fixed false timing	2020 pre-entry cutoff	0.0037	0.0037	0.310
Fixed false timing	2021 pre-entry cutoff	0.0051	0.0040	0.199
Country-label permutation	14 placebo countries, 499 draws	-0.0103	0.0350	0.774
Mixed country-date permutation	14 placebo countries, 499 draws	-0.0056	0.0160	0.407

Notes: Timing rows report pair-clustered standard errors. Spatial and mixed rows summarize 499 permutations; the reference p is the two-sided share of placebo coefficients at least as large in absolute value as the observed estimate. All models include relationship-pair and year fixed effects. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

4.3.3. Alternative treatment timing, samples, and estimator

Table 5 shows that the pair-clustered estimate remains positive and statistically significant when treatment is assigned using member-specific dates (0.0144), 2020–2021 observations are removed (0.0151), the sample begins in 2019 (0.0107), partners from the largest non-RCEP comparison market are excluded (0.0098), or offshore financial centers are excluded (0.0171). The exclusions reduce reliance on comparison markets that may have distinctive

post-2022 dynamics, and their positive estimates indicate that neither group alone drives the baseline result.

We also implement a double machine-learning sensitivity analysis. The outcome is the change in the relationship’s active status between the pre- and post-policy periods. Five-fold cross-fitting is grouped by partner country. Gradient-boosting regression and classification models estimate the outcome and treatment nuisance functions using pre-policy relationship history, firm characteristics, and trends. The orthogonalized estimate is 0.0077 (standard error 0.0021, $p = 0.002$; 95 percent confidence interval [0.0034, 0.0116]). Because this procedure uses a pre/post change rather than the annual fixed-effects outcome, it is a functional-form and high-dimensional-control check rather than a mechanically identical re-estimate of β .

Table 5: Robustness checks

Specification	Estimate	s.e.	p -value
Member-specific entry dates	0.0144***	0.0041	< 0.001
Exclude 2020–2021 observations	0.0151***	0.0047	0.001
2019–2024 sample window	0.0107**	0.0042	0.011
Exclude the largest non-RCEP comparison market	0.0098**	0.0042	0.019
Exclude offshore financial centers	0.0171***	0.0042	< 0.001
Double machine learning	0.0077***	0.0021	0.002

Notes: Standard errors are clustered by relationship pair. The double-machine-learning row uses five-fold cross-fitting and 999 pair-clustered bootstrap replications. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5. Heterogeneity analysis

The average estimate can conceal differences in how relationships respond to the agreement. We therefore examine the four heterogeneity predictions in H3a–H3d: supplier–customer direction, pre-policy relationship maturity, prior trade-agreement coverage, and manufacturing exposure. The pooled specifications in Table 6 estimate group-specific treatment coefficients in a common regression and test their equality directly.

For a partition indicator G_{ijr} , the pooled heterogeneity specification is

$$Active_{ijrt} = \beta_1 D_{jt}(1 - G_{ijr}) + \beta_2 D_{jt}G_{ijr} + \alpha_{ijr} + \lambda_t + \varepsilon_{ijrt}, \quad (9)$$

and the reported contrast is $\beta_2 - \beta_1$. Estimating both group terms in the same regression ensures that the comparison uses a common set of fixed effects and observations.

5.1. Relationship direction

Supplier relationships, which correspond to Chinese firms purchasing from foreign partners, have an estimated effect of 0.0295 ($p < 0.001$), while customer relationships have an estimate of -0.0033 ($p = 0.531$). The difference is 0.0327 and is statistically significant at the 1% level. The concentration of the response in supplier links is consistent with regional cumulation operating most directly through intermediate-input procurement and supports H3a.

5.2. Pre-policy relationship maturity

The scope for policy-induced adjustment may also depend on relationship-specific capital accumulated before implementation. Long-standing links have already absorbed search, screening, negotiation, and adaptation costs, leaving less room for an institutional change to alter their annual activity. We therefore classify a relationship as long-standing when it was active at year-end 2021 and had accumulated at least three years of relationship age by that date. The three-year threshold corresponds to the complete 2019–2021 pre-policy window and is fixed independently of the estimated coefficients. The remaining disclosed pairs form the comparison group. Both group-specific effects are estimated jointly in the baseline specification. The estimate for other relationships is 0.0183 ($p < 0.001$), whereas the estimate for long-standing relationships is -0.0162 and statistically insignificant ($p = 0.309$). The difference is -0.0345 ($p = 0.036$), indicating that the positive response is concentrated among relationships with greater scope for adjustment rather than among mature incumbent links, consistent with H3b.

5.3. Prior trade-agreement coverage

RCEP did not provide the same degree of institutional novelty across members. China already had free-trade agreements with the ASEAN members, Australia, New Zealand, and Korea, whereas RCEP created the first free-trade agreement between China and Japan. We therefore compare Japan with the RCEP partners covered by an earlier agreement, using the same set of non-RCEP relationships as the common comparison group. The coefficient for partners with prior agreement coverage is -0.0026 ($p = 0.605$), while the coefficient for Japan is 0.0378 ($p < 0.001$). Their difference is 0.0403 ($p < 0.001$). This pattern is consistent with a larger initial response where RCEP introduced genuinely new bilateral market-access rules and supports

H3c, although the Japan estimate also reflects characteristics specific to a single partner economy.

5.4. *Manufacturing exposure among supplier links*

Regional cumulation and origin requirements are most directly relevant to firms that organize manufacturing inputs across borders. We classify firms using the latest primary-industry code observed no later than 2021 and restrict this comparison to supplier relationships. The estimated effect is 0.0274 for manufacturing firms ($p = 0.030$), compared with -0.0205 for non-manufacturing firms ($p = 0.263$). The difference is 0.0479 ($p = 0.031$). The supplier-side response is therefore concentrated among manufacturing firms, supporting H3d and reinforcing the input-sourcing rationale for the directional asymmetry in H3a.

Table 6: Heterogeneity analysis

Partition	Group 1	Group 2	Difference (Group 2 – Group 1)
Relationship direction	Customer: -0.0033 (0.0052)	Supplier: 0.0295*** (0.0063)	0.0327*** (0.0082)
Pre-policy relationship maturity	Other: 0.0183*** (0.0041)	Long-standing: -0.0162 (0.0160)	-0.0345** (0.0165)
Pre-RCEP FTA coverage	Existing FTA: -0.0026 (0.0050)	Japan: 0.0378*** (0.0061)	0.0403*** (0.0075)
Firm sector, supplier links	Non-manufacturing: -0.0205 (0.0183)	Manufacturing: 0.0274** (0.0127)	0.0479** (0.0223)

Notes: Group coefficients are estimated jointly. A long-standing relationship is active at year-end 2021 and is at least three years old. The sector comparison is restricted to supplier links with pre-policy industry data. All models include relationship-pair and year fixed effects; relationship-pair clustered standard errors are in parentheses. The difference is Group 2 minus Group 1. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6. Further analysis

6.1. *Pre-existing external tariff exposure*

Because RCEP can generate trade diversion and production-network spillovers outside member economies, a separate alternative explanation is that the post-2022 pattern reflects firms' reorientation following pre-existing tariff pressure outside the agreement. To assess this possibility, we interact RCEP exposure with a firm-level measure of additional-duty exposure fixed in 2019 and estimate Eq. (11):

The firm-level exposure measure is constructed before the policy period as the fixed-weight average of product-level additional duties,

$$TariffExposure_i^{2019} = \sum_h \omega_{ih,2016} AdditionalDuty_h^{2019}, \quad \sum_h \omega_{ih,2016} = 1, \quad (10)$$

where h indexes products and $\omega_{ih,2016}$ are the firm's pre-policy export weights. This construction prevents post-entry trade shares from entering the exposure measure.

$$\begin{aligned} Active_{ijrt} = & \beta_1(Post2022_t \times RCEP_j) + \beta_2(Post2022_t \times TariffExposure_i) \\ & + \beta_3(Post2022_t \times RCEP_j \times TariffExposure_i) \\ & + \alpha_{ijr} + \lambda_t + \varepsilon_{ijrt}. \end{aligned} \quad (11)$$

The product weights underlying the exposure measure are fixed in 2016, and the firm-level measure is fixed before RCEP entered into force. A positive β_3 would indicate stronger reorientation toward RCEP partners among firms with greater pre-existing exposure.

Table 7 reports estimates on the subsample for which 2019 firm-level tariff exposure can be matched. The first row is therefore a matched-sample baseline, rather than a replication of the full-sample estimate in Table 2; the restriction leaves 253,184 observations and 31,648 relationship pairs. The matched-sample coefficient is 0.0061 (standard error 0.0065, $p = 0.346$) and changes little after adding the post-period tariff-exposure control (0.0063, standard error 0.0065, $p = 0.331$). The triple interaction is 0.0471 (standard error 0.0606, $p = 0.437$). Thus, firms with higher pre-existing tariff exposure do not exhibit a statistically larger differential change in RCEP relationships after 2022. The evidence does not support this external-tariff reorientation explanation, but the matched sample covers only 40.2 percent of relationship pairs and the imprecision of the estimates prevents a strict exclusion claim.

Table 7: Further analysis: Pre-existing external tariff exposure

Term	Estimate	s.e.	<i>p</i> -value
Post2022 $\times$ RCEP (matched baseline)	0.0061	0.0065	0.346
Post2022 $\times$ RCEP (matched + tariff control)	0.0063	0.0065	0.331
Post2022 $\times$ RCEP $\times$ tariff exposure	0.0471	0.0606	0.437

Notes: The sample is restricted to firm–partner relationships for which the 2019 firm-level tariff exposure can be matched (253,184 observations and 31,648 relationship pairs). All models include relationship-pair and year fixed effects, with standard errors clustered by relationship pair. The triple interaction tests whether high-exposure firms experienced an additional change in relationships with RCEP partners.

7. Conclusion and policy recommendations

This paper examines how RCEP’s entry into force relates to the international supply-chain relationships of Chinese listed firms. In a panel of 630,280 relationship–year observations and 78,785 relationship pairs, the preferred specification yields a 0.0136 differential increase in the probability that an RCEP relationship is active at year-end. The corresponding estimate is 0.0144 under member-specific implementation dates. The response is concentrated in supplier relationships, whose estimated increase is 0.0295, compared with an insignificant estimate of -0.0033 for customer relationships. It is also larger for Japan than for members with prior free-trade-agreement coverage (a difference of 0.0403) and, among supplier links, for manufacturing than for non-manufacturing firms (a difference of 0.0479). At the partner-country–year level, the bootstrap indirect associations are 0.0150 for incremental tariff relief (95% CI [0.0087, 0.0209]), 0.0003 for the intermediate-goods share of bilateral imports (95% CI [0.0002, 0.0004]), and 0.0274 through a lower partner-country import share (95% CI [0.0240, 0.0307]). These aggregate patterns, together with the supplier-side response, are consistent with RCEP strengthening regional production-network integration and thereby supporting the maintenance or formation of disclosed supplier relationships.

The findings carry three policy implications. First, implementation efforts should give firms clear and accessible guidance on regional cumulation, tariff schedules, and documentary requirements. The value of a common agreement depends not only on formal concessions but also on whether firms can apply the rules at reasonable administrative cost. Second, the increase in the member-level intermediate-goods share, together with the supplier-side

relationship response, suggests that trade-facilitation policies should support the matching of firms with qualified regional suppliers. Third, policymakers should monitor whether the early expansion of regional links reduces dependence on individual sources while preserving resilience to disruptions. These measures would help translate formal integration into durable firm-level relationships.

The evidence should nevertheless be interpreted within the limits of the design. The joint Wald test of the four pre-treatment event-study coefficients does not reject at the 5% level but is marginal at the 10% level ($\chi^2(4) = 8.389$, $p = 0.078$). The non-RCEP comparison group may itself respond to trade diversion or production-network spillovers generated by RCEP and is therefore not guaranteed to provide a completely untreated counterfactual. The estimates remain positive after excluding the largest non-RCEP comparison market and offshore financial centers. FactSet Revere primarily covers listed and large firms and does not report transaction values, customs declarations, certificates of origin, or firms' actual preference utilization. Moreover, the 2022–2024 period captures only the initial phase of a tariff schedule extending over several years. Linking product-level customs records to firm identifiers and extending the post-entry window would allow future research to distinguish relationship formation, survival, and trade-value responses more precisely.

References

- Ali, W., Saini, S., Dash, D.P., 2026. Institutional governance and energy security: Unveiling the drivers of global value chain participation in Asian economies. *Journal of Asian Economics* 103, 102108.
- Baier, S.L., Bergstrand, J.H., 2007. Do free trade agreements actually increase members' international trade? *Journal of International Economics* 71(1), 72–95.
- Croignani, M., Macchiavelli, M., Silva, A.F., 2022. Pirates without borders: The propagation of cyberattacks through firms' supply chains. *Journal of Financial Economics* 147(2), 432–456.
- Du, Y., Yan, J., Cao, F., Li, Y., Zhou, M., 2023. Higher education expansion and domestic value added in exports: Theory and evidence from China. *Journal of Asian Economics* 87, 101621.

- Garg, S., Ghosh, P., Tan, B.J., 2023. Within firm supply chains: Evidence from India. *Journal of International Economics* 144, 103793.
- Grossman, G.M., Helpman, E., Redding, S.J., 2024. When tariffs disrupt global supply chains. *American Economic Review* 114(4), 988–1029.
- Handley, K., Limão, N., Ludema, R.D., Yu, Z., 2024. Firm input choice under trade policy uncertainty. *Journal of International Economics* 150, 103909.
- Hayakawa, K., 2022. Multiple preference regimes and rules of origin. *Review of World Economics* 159(3), 673–696.
- Hayakawa, K., Laksanapanyakul, N., Yoshimi, T., 2023. Firm-level utilization rates of regional trade agreements: Importers’ perspective. *Journal of Asian Economics* 86, 101610.
- Hayakawa, K., Yamanouchi, K., 2023. How does the reform of rules of origin affect firm performance in importing countries? *The World Economy* 47(2), 837–856.
- Hayakawa, K., Kimura, F., Mukunoki, H., Urata, S., 2024. Negotiating over the rules of origin in regional trade agreements in Asia. *The World Economy* 47(7), 3097–3119.
- He, Y., Huo, W., Yu, J., 2023. Tracing the regional dual value chains: Measurement on the production position and evidence from China. *Journal of Asian Economics* 85, 101591.
- Head, K., Mayer, T., Melitz, M., 2024. The Laffer curve for rules of origin. *Journal of International Economics* 150, 103911.
- Huang, F., Li, J., Shen, J., Wang, H., 2026. Digital economy development and repositioning in the global intermediate goods trade network. *Journal of Asian Economics* 105, 102218.
- Kiyota, K., 2022. The COVID-19 pandemic and the world trade network. *Journal of Asian Economics* 78, 101419.
- Lee, W., Mulabdic, A., Ruta, M., 2023. Third-country effects of regional trade agreements: A firm-level analysis. *Journal of International Economics* 140, 103688.

- Li, Y., Wan, Y., Xie, S., 2025. The employment effects of regional market integration: Evidence from China. *Journal of Asian Economics* 101, 102053.
- Li, N., Li, R., 2026. A study on the impact of regional trade agreement digital trade rules on global innovation network. *Journal of Asian Economics* 103, 102144.
- Liu, Y., Deng, X., Min, J.H., Wang, J., Liu, Y., 2026. Multidimensional risks and the cleaner production-oriented trade transformation in the China–ASEAN. *Journal of Asian Economics* 103, 102128.
- Liu, B., Zhen, Y., 2022. Trade and rule effects of RCEP. *World Economy Studies*.
- Ma, S., Huang, S., Wu, P., 2024a. Data policy restrictions and cross-border e-commerce: Evidence from China. *Journal of Asian Economics* 95, 101826.
- Ma, S., Ren, F., Feng, F., Feng, K., 2024b. Cross-border personnel mobility and bilateral value chain linkages: Evidence from visa liberalization in China. *Journal of Asian Economics* 95, 101844.
- Narayan, P.K., Rath, B.N., Syarifuddin, F., 2022. Understanding the role of trade agreements in Indonesia’s FDI. *Journal of Asian Economics* 82, 101532.
- Newman, C., Rand, J., Tarp, F., 2023. Imports, supply chains and firm productivity. *World Development* 172, 106371.
- Ni, S., Bai, X., Li, L., 2022. Behind the heterogeneous trade effects of standards: Multi-sector evidence from China. *Journal of Asian Economics* 82, 101513.
- Ou, J., Yang, T., Yao, S., Hu, M., 2026. Digital finance, supply chain spillovers and corporate innovation: Evidence from Chinese listed firms 2011–22. *Journal of Asian Economics* 104, 102175.
- Peng, Q., Chen, Z., Li, J., 2025. Financing constraints and the export partnerships with RCEP: Evidence from manufacturing firms in China. *Journal of Asian Economics* 97, 101870.
- Ping, L., 2021. Institutional characteristics and economic effects of RCEP. *Asia-Pacific Economic Review*.

- Poilly, C., Tripier, F., 2025. Regional trade policy uncertainty. *Journal of International Economics* 155, 104078.
- Shen, M., Li, T., 2020. RCEP: Institutional framework, rules, and regional effects. *International Economic Review*.
- Sun, J., Luo, Y., Zhou, Y., 2022. The impact of regional trade agreements on the quality of export products in China's manufacturing industry. *Journal of Asian Economics* 80, 101456.
- Sun, K., Xiao, H., Jia, Z., Tang, B., 2023. Estimating the effects of regional value chains of the RCEP in a GVC-CGE model. *Journal of Asian Economics* 88, 101647.
- Tanaka, K., Fukunishi, T., 2022. Rules of origin and exports in developing economies: The case of garment products. *Journal of Asian Economics* 82, 101514.
- Wu, J., Bian, Y., Bai, J., 2023. Global value chain upgrading effect of foreign capital withdrawal: Evidence from Chinese firms. *Journal of Asian Economics* 89, 101672.
- Xie, D., 2023. Tariff cost and cross-border M&A affiliate sales: Evidence from China. *Journal of Asian Economics* 87, 101636.
- Xu, L., Jing, J., Wu, C., 2024. Impact of financial development on the position in global value chain: An analysis from the perspective of R&D intensity. *Journal of Asian Economics* 92, 101742.
- Zhang, H., Ding, Y., 2024. The impact of global value chain restructuring on the product structure of China–EU exports. *Journal of Asian Economics* 91, 101708.
- Zhou, F., Wen, H., 2022. Trade policy uncertainty, development strategy, and export behavior: Evidence from listed industrial companies in China. *Journal of Asian Economics* 82, 101528.
- Zhou, H., Wu, L., 2025. Rules of origin, import of inputs and firm innovation. *Journal of Asian Economics* 96, 101872.

- Zhu, J., Liu, X., Cao, L., 2025. The Belt and Road Initiative and China's diversifying food imports: A perspective on the "Five-pronged approach." *Journal of Asian Economics* 96, 101849.
- Zhu, Y., Sun, C., 2025. Environmental provisions in preferential trade agreements and export product quality of Chinese firms. *Journal of Asian Economics* 101, 102048.